MILITARY SPECIFICATION SHEET

Ⓔ RELAYS, SOLID STATE, TRACK MOUNTED, ELECTRONIC TIME DELAY ON RELEASE, TYPE IIA, CLASS C, SPST, 150 MILLIAMPERES, FIXED TIME, 50 MILLISECONDS TO 500 SECONDS

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of of MIL-R-83726.

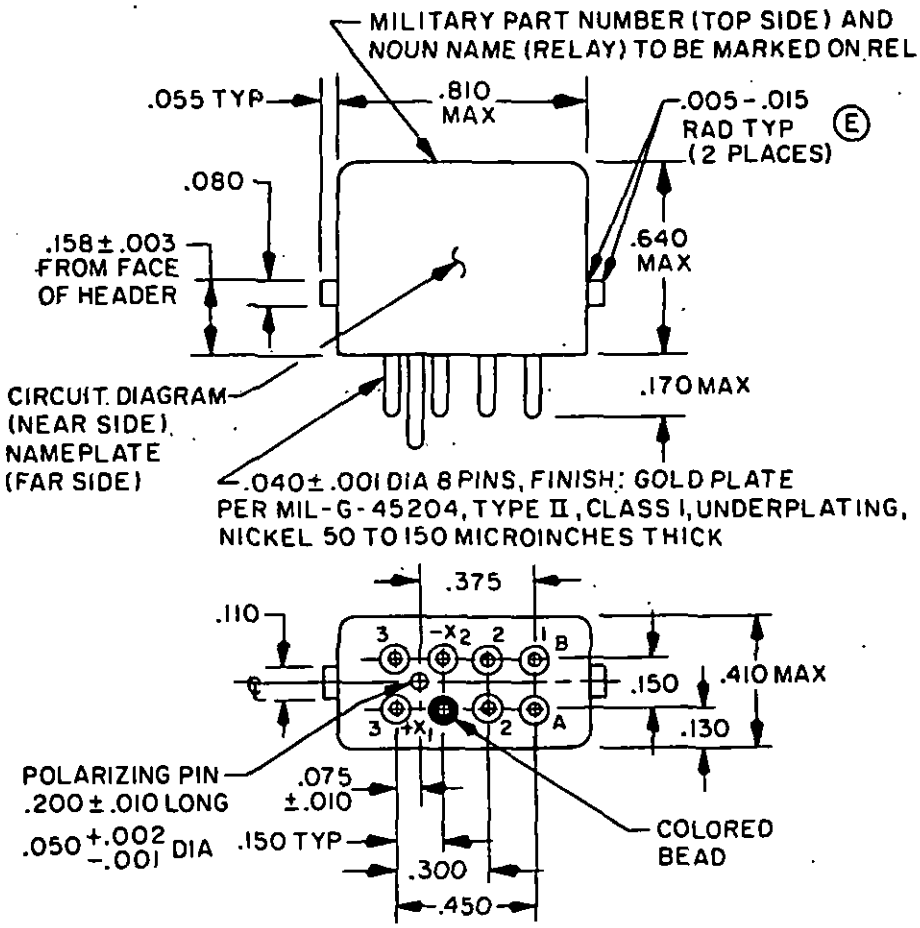
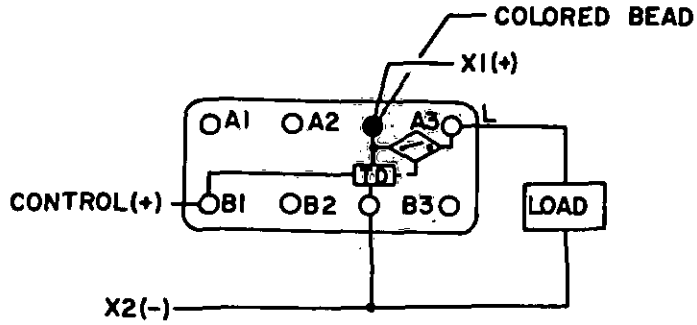


FIGURE 1. Outline dimensions and configuration.

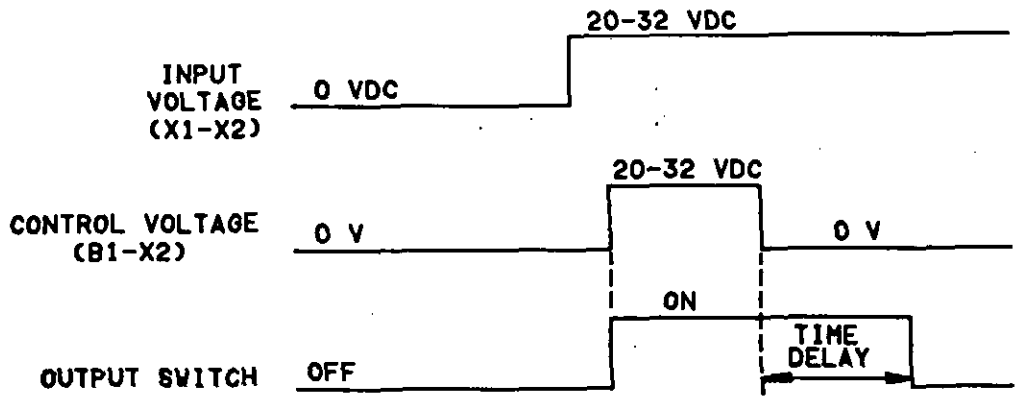
Ⓔ denotes change

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CIRCUIT DIAGRAM
(SEE NOTES 4 AND 5)



TIMING DIAGRAM

INCHES	MM	INCHES	MM
.001	0.03	.080	2.03
.002	0.05	.110	2.79
.003	0.08	.130	3.30
.005	0.13	.150	3.81
.010	0.25	.158	4.01
.015	0.38	.170	4.32
.040	1.02	.200	5.08
.050	1.27	.410	10.41
.055	1.40	.640	16.26
.075	1.90	.810	20.57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Terminals A1, A2, B1 and B2 are connected internally. Do not use for external tie points or for terminals.
5. There shall be affixed to the relay a permanent legible circuit diagram that identifies each terminal location specified.
6. Relays are intended for use with relay socket mount MIL-R-6106/23 and relay bracket mount MIL-R-6106/25.

Ⓔ FIGURE 1. Outline dimensions and configuration - Continued.

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REQUIREMENTS:

OPERATING DATA:

Timing action: Delay-on release.

Time delay: Fixed; select from .05 to 500 seconds.

Timing accuracy 1/: ±10 percent of the nominal value.

Recycle time 2/: 10 milliseconds.

Power interrupt: The accuracy will not be affected by power interruptions up to 1 millisecond spaced at least 10 milliseconds apart.

Control voltage (81-X2): 20 to 32 V dc.

Control current: 2 milliamperes maximum.

INPUT DATA:

Input voltage: 28 V dc nominal; range 20 to 32 V dc.

Duty rating: Continuous.

Current drain: 10 milliamperes maximum plus load at 25°C.

Polarity protection: The timer shall be inoperative during, and undamaged by, reversal of the polarity of the operating voltage.

OUTPUT DATA:

Configuration: SPST; switch closure to +28 V dc.

Rating: 150 milliamperes maximum.

Suppression: Inductive suppression provided for output protection.

Voltage drop: 2 V dc maximum.

Leakage current: 1.0 microampere maximum at 28 V dc and 25°C; 10 microamperes maximum at 28 V dc and 125°C.

Ⓔ Endurance (life): 1,000,000 cycles minimum for test.

Qualification: 2,000 hours.

Group B: 1,000 hours or 1,000,000 cycles, whichever is less.

1/ The accuracy requirement applies for any combination of operating temperature and voltage.

Ⓔ 2/ Recycle time is defined as the minimum time that power must be applied to the control terminal to assure that the next timing cycle will be completed within the specified timing tolerance. (Units can be recycled during timing or after time-out.)

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ELECTRICAL DATA:

Transients 3/:

Positive transients: +80 volts per MIL-STD-704A, figure 9, limit 1.

Self-generated transients: ±50 volts maximum.

Spike susceptibility: ±600 volts, 10 microseconds maximum.

Electromagnetic interference 4/: Per MIL-STD-461, class 1D.

Insulation resistance 5/: 1,000 megohms at 500 V dc between each pin and case.

Dielectric strength 5/: 1,000 volts rms at 60 hertz at sea level between case and pins connected together.

ENVIRONMENTAL DATA:

Ambient temperature (operating and nonoperating): -55°C to +125°C.

Vibration (sinusoidal): 10 to 55 hertz at 0.06 inch DA; 10 g, 55 to 500 hertz; 20 g, 500 to 3,000 hertz.

Vibration (random): MIL-STD-202, method 214, condition I, letter C, duration of 30 minutes.

Shock: 1,100 g for 0.5 millisecond, 1/2 sine, 3 axis.

Ⓔ Acceleration: 100 g, 3 axis.

Ⓔ Seal: Hermetic.

Moisture resistance: MIL-STD-202, method 106.

PHYSICAL DATA:

Dimensions and configuration: See figure 1.

Terminations: See figure 1.

Terminal strength: 3 pounds pull.

Weight: 0.56 ounce.

Marking: Military part number. Circuit diagram. Manufacturer's part number and name, date code.

Part number: Consists of the basic number of this specification sheet and a dash number denoting the required time delay, expressed in milliseconds. The first three digits are significant figures. The last digit is the number of zeros following the first three. A final letter, if any, designates mounting type.

NOTE: Additional time delay relays within the 50-millisecond to 500-second delay range are available. Part number may be established as follows:

Examples: M83726/25-0500 equals a 50-millisecond time delay.
M83726/25-2503 equals a 250-second time delay.
M83726/25-5003 equals a 500-second time delay.

3/ Transient specifications are based on a maximum duty cycle of 1/50.

4/ EMI test limits will not be exceeded during the timing interval or when continuously energized under steady-state conditions per Radio Interference requirements of MIL-R-83726.

5/ Terminals X1, X2 and B1 and A3 must be connected together during this test. Dielectric withstanding voltage and insulation resistance are measured between all mutually insulated terminals and case.

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Preparing activity:
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